

● FLAGSHIP PROGRAM · 14 WEEKS

AI Engineering & Generative AI Program

the complete path to a hireable AI engineer.

A 3.5-month, project-driven program that takes you from Python fundamentals to production-ready AI systems. Go deep on machine learning, LLMs, prompt engineering, embeddings, vector databases, RAG, AI agents and MCP servers — then ship a complete, deployable AI product and a professional GitHub portfolio.

[Apply for this track →](#)[Book Free Counselling](#)**14****Weeks**

3.5 MONTHS

6**Mini-Projects**

PLUS CAPSTONE

8**Modules**

STRUCTURED PATH

100%**Portfolio**

GITHUB-READY

RECOMMENDED FORMAT

5 days per week

2–3 hours per day

60% practical coding

20% concepts & architecture

20% projects & code reviews

Weekly coding assignments

6 mini-projects

1 major capstone project

GitHub portfolio development

Mock interviews & presentations

01

WEEKS 1–2

MODULE

Python & Software Engineering Foundations

Build the programming and engineering foundation required to develop production-ready AI applications.

Week 1 — Python for AI Engineering

- Python environment setup
- VS Code & development workflow
- Variables and data types
- Strings & string manipulation
- Lists, tuples, sets & dictionaries
- Conditional statements
- Loops and iterations
- Functions
- Lambda functions
- List & dictionary comprehensions
- Exception handling
- File handling
- Working with JSON and CSV files

PRACTICAL EXERCISES

- Student marks analyser
- Expense tracker
- CSV data processor
- JSON parser
- Log file analyser

WEEKLY ASSIGNMENT

Build a command-line application that reads data from CSV or JSON files, processes the information, and generates a report.

Week 2 — Professional Python Development

- Object-oriented programming
- Classes and objects
- Inheritance
- Composition
- Modules and packages
- Type hints
- Pydantic models
- Environment variables
- Virtual environments
- pip & package management
- Clean code principles
- Debugging techniques
- Logging
- Unit testing basics
- Git fundamentals
- GitHub workflow
- Branching & pull requests
- REST API fundamentals
- HTTP methods
- Authentication
- Working with external APIs
- Building APIs with FastAPI

MINI PROJECT 1 • AI-READY PYTHON API SERVICE

- Connects to an external API
- Processes API responses
- Validates data using Pydantic
- Provides REST endpoints
- Includes error handling
- Uses environment variables
- Includes basic unit tests
- Is published to GitHub

02

WEEKS 3-4

MODULE

Data, Machine Learning & Neural Network Foundations

Understand the core concepts behind machine learning and modern AI systems without getting lost in unnecessary mathematics.

Week 3 — Data Engineering for AI

- NumPy fundamentals
- Pandas fundamentals
- DataFrames & Series
- Loading CSV, JSON & API data
- Data cleaning
- Missing values
- Data transformation
- Filtering and grouping
- Feature preparation
- Exploratory data analysis
- Data visualisation basics
- Understanding datasets
- Training, validation & test datasets

PRACTICAL EXERCISES

- Clean a real-world dataset
- Customer data analysis
- Sales prediction dataset preparation
- Data quality report generation

WEEKLY ASSIGNMENT

Build an automated data-analysis application that accepts a CSV file and generates summary statistics and insights.

Week 4 — Machine Learning & Deep Learning Essentials

- What is artificial intelligence?
- AI vs ML vs DL vs generative AI
- Supervised learning
- Unsupervised learning
- Classification
- Regression
- Clustering
- Feature engineering
- Model training
- Overfitting and underfitting
- Accuracy, precision, recall & F1
- Introduction to scikit-learn
- Neural network fundamentals
- Neurons, layers, weights & activations
- Training and inference
- Transformers introduction
- Attention mechanism intuition
- Why transformers changed AI

PRACTICAL LABS

- Spam classifier
- Customer churn prediction
- Simple recommendation engine
- Sentiment classifier

MINI PROJECT 2 · INTELLIGENT PREDICTION APPLICATION

- Dataset preparation
- Feature engineering
- Model training
- Model evaluation
- Prediction API
- Simple frontend or API interface

03

WEEKS 5-6

MODULE

Large Language Models & Prompt Engineering

Understand how modern LLMs work and learn to build reliable applications using commercial and open-source AI models.

Week 5 — LLM Fundamentals

- What is a large language model?
- Transformer architecture overview
- Tokens and tokenisation
- Context windows
- Embeddings
- Training vs fine-tuning vs inference
- Temperature and sampling
- Top-p
- System prompts
- User prompts
- Assistant messages
- Model limitations
- Hallucinations
- Context limitations
- Cost & latency considerations
- Hosted models vs local models
- Introduction to major AI model APIs

API LABS

- Text generation APIs
- Chat completion APIs
- Streaming responses
- Multimodal inputs
- Token usage tracking
- Error handling
- Retry mechanisms
- Rate-limit handling

PRACTICAL APPLICATIONS

- AI content generator
- Email assistant
- Text summariser
- Sentiment analyser
- Translation assistant

Week 6 — Advanced Prompt Engineering

- Zero-shot prompting
- Few-shot prompting
- Role prompting
- Context engineering
- Chain-of-thought concepts
- Prompt decomposition
- Prompt chaining
- Structured outputs
- JSON schema outputs
- Tool calling
- Function calling
- Prompt templates
- Dynamic prompts
- Prompt injection
- Prompt security
- Evaluating prompt quality
- Building reusable prompt libraries

PRACTICAL LABS

- Resume analyser
- Invoice information extractor
- Customer-support classifier
- Product description generator
- Meeting-note summariser
- Structured data extraction system

MINI PROJECT 3 • LLM-POWERED BUSINESS APPLICATION

- Choose one: HR / sales / support / content / document / interview assistant
- Model API integration
- Structured output
- Prompt templates
- Error handling
- Response validation
- Basic evaluation

04

WEEKS 7-8

MODULE

Embeddings, Vector Databases & Semantic Search

Learn how AI applications understand meaning and search through large amounts of unstructured information.

Week 7 — Embeddings & Semantic Search

- What are embeddings?
- Vector representations
- Similarity search
- Cosine similarity
- Keyword search vs semantic search
- Creating embeddings
- Storing embeddings
- Metadata filtering
- Hybrid search
- Reranking concepts

PRACTICAL LABS

- Similar movie finder
- Semantic FAQ search
- Product recommendation search
- Resume-to-job matching
- Knowledge-base search

Week 8 — Vector Databases & AI Application Frameworks

- Vector database architecture
- Chroma
- pgvector
- Pinecone concepts
- Collections and indexes
- Metadata
- Similarity search
- Filtering
- Document ingestion
- Document loaders
- Text splitters
- Chunking
- Embedding pipelines
- Introduction to LangChain
- Chains
- Prompt templates
- Model components
- Output parsers
- Runnable pipelines
- Conversation history

MINI PROJECT 4 • SEMANTIC SEARCH APPLICATION

- Pick a domain: movie / catalogue / jobs / knowledge / products / legal
- Document ingestion & chunking
- Embedding generation
- Vector storage
- Semantic search
- Metadata filtering
- Result ranking
- Web interface

05

WEEKS 9-10

MODULE

Retrieval-Augmented Generation — RAG

Build AI systems capable of answering questions from private company data and documents.

Week 9 — RAG Fundamentals

- Why RAG is needed
- RAG architecture
- Data ingestion
- Document processing
- PDF extraction
- Website ingestion
- CSV & database ingestion
- Chunking strategies
- Fixed-size chunking
- Recursive chunking
- Semantic chunking concepts
- Embedding generation
- Retrieval
- Context construction
- Answer generation
- Source citation

PRACTICAL LAB

Build a basic PDF question-answering application.

Week 10 — Advanced RAG

- Query transformation
- Multi-query retrieval
- Hybrid retrieval
- Metadata filtering
- Reranking
- Context compression
- Conversation-aware retrieval
- RAG memory
- Citation generation
- Hallucination reduction
- Retrieval evaluation
- Answer evaluation
- Faithfulness
- Relevance
- Latency optimisation
- Cost optimisation
- RAG security considerations

MINI PROJECT 5 • ENTERPRISE KNOWLEDGE ASSISTANT

- Answers from PDFs, Word docs, web pages, FAQs & internal knowledge
- Document upload & automatic indexing
- Semantic retrieval
- Conversational interface with history
- Source citations
- Hallucination control
- Feedback mechanism
- Basic evaluation dashboard

06

WEEKS 11–12

AI Agents, Tools & Workflow Automation

MODULE

Move from simple chatbots to AI systems capable of reasoning about tasks, selecting tools, and completing multi-step workflows.

Week 11 — AI Agent Fundamentals

- What is an AI agent?
- Chatbot vs workflow vs agent
- Agent loop
- Reasoning and action
- Tool selection
- Function calling
- Agent state
- Short-term memory
- Long-term memory concepts
- Planning
- Reflection concepts
- Human-in-the-loop workflows
- Agent guardrails
- Error recovery
- Retry strategies
- Deterministic vs agentic workflows

PRACTICAL AGENTS

- Web research assistant
- Database query agent
- Email drafting assistant
- Customer-support agent
- Data analysis agent
- Task planning agent

Week 12 — LangGraph & Multi-Step AI Systems

- Graph-based AI workflows
- Nodes and edges
- State management
- Conditional routing
- Tool nodes
- Human approval steps
- Persistence
- Checkpointing
- Error handling
- Multi-agent architecture
- Supervisor agents
- Specialist agents
- Agent communication
- When not to use multi-agent systems

MINI PROJECT 6 • MULTI-TOOL AI AGENT

- Understands user requests
- Selects the appropriate tool
- Searches a knowledge base
- Queries an API
- Performs calculations
- Maintains conversation state
- Asks for human approval where required
- Returns structured results

07

WEEK 13

MCP Servers & AI Integrations

MODULE

Learn how AI applications connect securely to external tools, APIs, databases, and business systems.

Week 13 — Model Context Protocol

- What is Model Context Protocol?
- MCP architecture
- MCP clients and servers
- Tools
- Resources
- Prompts
- Building an MCP server
- Connecting REST APIs
- Connecting databases
- Authentication concepts
- Permission design
- Tool descriptions
- Input validation
- Error handling
- Security considerations
- MCP testing and debugging

PRACTICAL LABS • BUILD MCP SERVERS FOR

- Product catalogue search
- CRM data access
- Company knowledge search
- Task management
- Database querying

WEEK 13 PROJECT • BUSINESS MCP SERVER

- Expose at least three useful business capabilities
- Example: e-commerce MCP server — search products
- Check inventory
- Retrieve order status
- Generate product recommendations

08

WEEK 14

Production AI Engineering & Capstone Project

MODULE

Bring everything together into a deployable AI product.

Week 14 — Production AI Topics

- AI application architecture
- Frontend & backend separation
- FastAPI AI backend
- Streaming responses
- Authentication
- API security
- Secrets management
- Rate limiting
- Caching
- Logging
- Observability
- Token tracking
- Cost tracking
- Prompt versioning
- Model fallback strategies
- AI response evaluation
- User feedback collection
- Guardrails
- Basic deployment concepts
- Docker fundamentals
- Cloud deployment overview

Build one complete **AI product**

Work individually or in teams to design, build and ship a production-oriented AI product. Choose one of the suggested tracks below.

AI Recruitment Platform 01

- Resume parsing
- Candidate scoring
- Semantic job matching
- AI interview questions
- Candidate summarisation
- Recruiter assistant

AI Customer Support Platform 02

- Company knowledge RAG
- Ticket classification
- Automated response suggestions
- Tool-using support agent
- Escalation workflow
- Conversation history

AI Sales Assistant 03

- Lead research
- Company analysis
- Personalised outreach generation
- CRM integration
- Follow-up recommendations
- Sales knowledge assistant

AI Education Tutor 04

- Course material ingestion
- RAG-based tutoring
- Quiz generation
- Student progress tracking
- Personalised explanations
- Learning recommendations

AI Jewellery Sales Assistant 05

- Semantic jewellery search
- Natural-language product discovery
- Product recommendations
- Jewellery knowledge RAG
- Customer preference understanding
- Sales-agent workflow

AI Movie Discovery Platform 06

- Natural-language movie search
- Semantic recommendation
- Movie knowledge assistant
- Personalised watchlist suggestions
- Review summarisation
- Conversational movie discovery

FINAL WEEK

Deliverables

✓ Working AI application

✓ GitHub repository

✓ Professional README

✓ Architecture diagram

✓ API documentation

✓ Demo video

✓ Deployment

✓ Project presentation

✓ Technical interview on architecture & decisions

TOOLS & TECHNOLOGIES

Everything you'll master

Programming

Python

SQL basics

JSON

REST APIs

Backend Development

FastAPI

Pydantic

Async API concepts

Data & Machine Learning

NumPy

Pandas

scikit-learn

Generative AI

Commercial LLM APIs

Open-source model concepts

Prompt engineering

Structured output

Function calling

AI Application Development

LangChain

LangGraph

Embeddings

Vector databases

RAG

AI agents

MCP servers

Vector Storage

Chroma

pgvector

Pinecone concepts

Engineering

Git

GitHub

Docker fundamentals

Environment management

Logging

Testing

API security

Deployment fundamentals

WEEKLY STRUCTURE

A repeatable rhythm

Monday

Concept introduction & instructor demonstration.

Tuesday

Guided coding lab.

Wednesday

Feature-building workshop.

Thursday

Project implementation & debugging.

Friday

Code review, demo, assessment & interview prep.

ASSESSMENT MODEL

How you're graded

20%

Weekly Assignments

Short coding & implementation assignments.

30%

Mini Projects

Six practical AI projects built during the course.

35%

Capstone Project

A complete, production-oriented AI application.

10%

Technical Interview

Architecture, fundamentals, AI concepts & project.

5%

Participation & GitHub

Code quality, commits, docs & collaboration.

By graduation, you can...

- ✓ Write production-quality Python applications
- ✓ Integrate modern LLM APIs
- ✓ Generate structured AI outputs
- ✓ Build semantic search applications
- ✓ Build complete RAG pipelines
- ✓ Create stateful workflows with LangGraph
- ✓ Evaluate AI application quality
- ✓ Add guardrails & reliability controls
- ✓ Present a professional GitHub portfolio
- ✓ Build REST APIs
- ✓ Engineer reliable prompts
- ✓ Implement function & tool calling
- ✓ Work with embeddings & vector databases
- ✓ Develop tool-using AI agents
- ✓ Build MCP servers
- ✓ Reduce hallucinations
- ✓ Deploy an end-to-end AI application

The objective is not to produce students who only know how to call an LLM API. This program produces **junior AI application engineers** who understand software engineering, data pipelines, LLM integration, retrieval systems, agents, tool use, evaluation and deployment.